

Literature-implied status for the nuclear-adjoint remark in arXiv:2204.12583

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Status. This is a *literature-implied answer*, not a new proof. The source remark is in Aksoy–Thiong, *Schauder’s Theorem and s -Numbers*, arXiv:2204.12583, Section “Hutton’s Theorem Revisited” after the proof of Hutton’s theorem. It says that, because nuclear operators are compact, it is natural to ask how nuclearity of an operator T and of its adjoint T^* are related. The same remark recalls the positive theorem that if X^* has the approximation property and T^* is nuclear, then T is nuclear and $\|T\|_{\mathcal{N}} = \|T^*\|_{\mathcal{N}}$.

Supporting identification. Reinov’s paper *On linear operators with s -nuclear adjoints*: $0 < s \leq 1$, arXiv:1311.2270, gives the broader relationship. Lemma 2 there states that for $T \in L(X, Y)$,

$$\pi_Y T \in N_s(X, Y^{**}) \iff T^* \in N_s(Y^*, X^*),$$

and that $T \in N_s(X, Y)$ implies $T^* \in N_s(Y^*, X^*)$. Theorem 1 states that if $s \in (0, 1]$ and either X^* or Y^{***} has the approximation property of order s , then T^* being s -nuclear implies T is nuclear. For $s = 1$, this includes the usual nuclear-adjoint converse under the classical approximation property assumptions.

Negative side. The converse fails in general without those approximation-property hypotheses. Reinov’s Theorem 2 states that for each $r \in (2/3, 1]$ there are a Banach space Z_0 and an operator $T : Z_0^{**} \rightarrow Z_0$ such that Z_0^{**} has a Schauder basis, Z_0^{***} has AP_s for every $s < r$, $\pi_{Z_0} T \in N_r(Z_0^{**}, Z_0^{**})$, but

$$T \notin N_1(Z_0^{**}, Z_0).$$

Taking $r = 1$, Lemma 2 identifies $\pi_{Z_0} T \in N_1$ with nuclearity of the adjoint T^* . Thus there is a non-nuclear operator whose adjoint is nuclear. This supplies the negative general answer to the natural converse question.

Scope. This answers only the nuclearity-versus-adjoint-nuclearity issue raised in the remark. It does not address the source paper’s other themes: comparison of general s -numbers, Hutton-type equalities, or Q-compactness under approximation schemes. Reinov’s paper predates arXiv:2204.12583 and does not explicitly cite that source, so this packet belongs in `literature_implied_answers`, not `literature_already_answered`.

References

- [1] A. G. Aksoy and D. A. Thiong, *Schauder’s Theorem and s -Numbers*, arXiv:2204.12583.

- [2] O. I. Reinov, *On linear operators with s -nuclear adjoints: $0 < s \leq 1$* , arXiv:1311.2270.